

AVENTO

Event Management System

SOFTWARE TESTING DOCUMENTATION

Unit Testing • Integration Testing • User Acceptance Testing

Prepared as part of the Software Development Life Cycle (SDLC)

July 11, 2026

1. Testing Overview

Testing is a crucial phase in the Software Development Life Cycle (SDLC) that ensures the developed application functions correctly, reliably, and efficiently according to the specified requirements. It helps identify defects, validate system behavior, and verify that all modules work together seamlessly while meeting user expectations.

For the AVENTO Event Management System, comprehensive testing was carried out through Unit Testing, Integration Testing, and User Acceptance Testing (UAT) to evaluate the functionality of individual modules, the interaction between integrated components, and the overall user experience. The successful completion of these testing activities confirmed that the application is stable, secure, responsive, and ready for deployment.

2. User Acceptance Testing (UAT)

2.1 Introduction

User Acceptance Testing (UAT) was conducted to evaluate the usability, functionality, and overall user experience of the AVENTO application. A group of students and organizers interacted with the system and provided valuable feedback. Based on their suggestions, improvements were implemented to enhance usability, interface consistency, and workflow efficiency.

2.2 User Acceptance Testing Table

Reviewer	Role	Feedback Received	Improvement Implemented	Status
User 1	Student	Navigation bar contained too many menu options.	Introduced My Workspace dropdown to group related pages.	**Passed**
User 2	Student	Password input did not allow users to verify typed passwords.	Added Show/Hide Password (eye icon) functionality.	**Passed**
User 3	Student	QR scanner required additional clicks to access the camera.	Scanner now opens the camera automatically after permission is granted.	**Passed**
User 4	Student Organizer	Wrong event QR codes could be scanned accidentally.	Added backend validation to verify event-specific QR codes.	**Passed**
User 5	Faculty	Certificates should only be generated for students who attended the event.	Restricted certificate generation to participants marked Present.	**Passed**
User 6	Student	QR passes remained visible after the event ended.	Automatically hides the View Pass option after event completion.	**Passed**
User 7	Student Organizer	Scan QR option should disappear after event completion.	Removed Scan QR button for completed events and blocked backend attendance requests.	**Passed**
User 8	Student	Dark mode appeared inconsistent across pages.	Redesigned the entire dark theme with a unified color palette.	**Passed**

Reviewer	Role	Feedback Received	Improvement Implemented	Status
User 9	Student	About page lacked creativity and branding.	Redesigned About page with storytelling, illustrations, and pastel-themed UI.	**Passed**
User 10	Student	Students could join multiple teams and organize the same event.	Added validation to prevent duplicate team participation and organizer conflicts.	**Passed**

2.3 UAT Summary

User Acceptance Testing confirmed that AVENTO provides an intuitive, responsive, and user-friendly experience. Feedback collected from users resulted in several enhancements to navigation, QR attendance, validation rules, UI consistency, and accessibility. After implementing these improvements, users confirmed that the system met their expectations and fulfilled the intended project objectives.

3. Unit Testing

3.1 Introduction

Unit Testing was performed to verify that each module of the AVENTO application functions correctly in isolation. Every module was tested independently before integration into the complete system.

3.2 Unit Testing Table

Test ID	Module	Test Description	Expected Result	Actual Result	Status
UT001	Authentication	Verify valid student login	Student dashboard opens successfully	Dashboard displayed	**Passed**
UT002	Authentication	Verify new user registration	User account created successfully	Account created	**Passed**
UT003	Event Management	Create a new event	Event stored successfully	Event created	**Passed**
UT004	Event Management	Edit event details	Updated information displayed	Event updated	**Passed**
UT005	Event Registration	Register for an individual event	Registration successful	Registration completed	**Passed**
UT006	Team Registration	Create a team	Team created successfully	Team created	**Passed**
UT007	Team Validation	Attempt duplicate team registration	Registration blocked	Validation successful	**Passed**

Test ID	Module	Test Description	Expected Result	Actual Result	Status
UT008	Organizer Validation	Team member requests organizer role	Request rejected	Validation successful	**Passed**
UT009	QR Pass	Generate QR pass	QR code generated correctly	QR displayed	**Passed**
UT010	QR Attendance	Scan valid QR code	Attendance marked Present	Database updated	**Passed**
UT011	QR Attendance	Scan duplicate QR	Duplicate attendance prevented	Validation successful	**Passed**
UT012	QR Attendance	Scan QR from another event	Request rejected	Validation successful	**Passed**
UT013	Certificate	Generate certificate	Certificate generated only for Present participants	Certificate created	**Passed**
UT014	Notifications	Trigger notification	Notification delivered	Notification received	**Passed**
UT015	Theme	Switch between Light/Dark mode	Theme changes successfully	Theme updated	**Passed**
UT016	Navigation	Open My Workspace dropdown	Dropdown functions correctly	Navigation successful	**Passed**
UT017	Event Lifecycle	Completed event validation	View Pass and Scan QR hidden	Validation successful	**Passed**
UT018	Logout	User logout	Session terminated successfully	Redirected to login	**Passed**

3.3 Unit Testing Summary

The results demonstrate that each functional unit performed according to the specified requirements. Authentication, event management, registration, QR attendance, certificate generation, notifications, navigation, and theme management were all tested individually and produced the expected outcomes.

4. Integration Testing

4.1 Introduction

Integration Testing was conducted to verify that the different modules of AVENTO communicate effectively and exchange data correctly. The objective was to ensure seamless interaction between the frontend, backend, and Supabase database.

4.2 Integration Testing Table

Test ID	Integrated Modules	Test Scenario	Expected Result	Actual Result	Status
IT001	Authentication + Dashboard	Login with valid credentials	User redirected to appropriate dashboard	Successful	**Passed**
IT002	Event Management + Database	Create event	Event stored and displayed to users	Successful	**Passed**
IT003	Registration + Database	Register for event	Registration reflected in My Registrations	Successful	**Passed**
IT004	Team Registration + Validation	Student joins a team	Team created and duplicate registration prevented	Successful	**Passed**
IT005	Organizer Recruitment + Validation	Team member applies as organizer	Request rejected according to rules	Successful	**Passed**
IT006	Registration + QR Pass	View event QR pass	QR generated using registration details	Successful	**Passed**
IT007	QR Scanner + Attendance + Database	Scan valid QR code	Attendance updated to Present	Successful	**Passed**
IT008	QR Scanner + Event Validation	Scan QR from different event	Attendance rejected	Successful	**Passed**
IT009	Attendance + Certificate	Generate certificate after attendance	Certificate generated successfully	Successful	**Passed**
IT010	Attendance + Certificate Validation	Generate certificate for absent student	Certificate generation blocked	Successful	**Passed**
IT011	Event Lifecycle + QR Module	Event reaches completion	View Pass and Scan QR removed automatically	Successful	**Passed**
IT012	Notifications + Registration	Register for event	Notification sent to student	Successful	**Passed**
IT013	Theme + UI Components	Toggle Light/Dark mode	Entire application theme updates consistently	Successful	**Passed**
IT014	Navigation + Routing	Access pages through My Workspace	Correct page navigation	Successful	**Passed**
IT015	Frontend + Backend + Supabase	Complete end-to-end event workflow	Data synchronized successfully across all layers	Successful	**Passed**

4.3 Integration Testing Summary

Integration Testing verified that all interconnected modules functioned seamlessly as a complete system. The interaction between the React frontend, Node.js/Express backend, and Supabase database was tested throughout the event lifecycle, including authentication, registrations, QR attendance, certificate generation, notifications, and validations. All integrated workflows executed successfully without affecting existing functionality.

5. Overall Testing Summary

The AVENTO Event Management System underwent comprehensive testing through Unit Testing, Integration Testing, and User Acceptance Testing. Each functional module was independently verified, interactions between integrated modules were validated, and feedback from end users was incorporated to improve usability and system reliability.

All test cases produced the expected outcomes, confirming that the application satisfies the functional requirements, maintains data consistency, and delivers a secure, responsive, and user-friendly experience suitable for deployment.